

SIMATS ENGINEERING
ASSESSMENT TOOL 3
MCQ Test

COURSE NAME: Computer Networks
COURSE CODE: CSA0708

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COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 20%

QUESTIONS:20

Total Marks:20

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?
 - a) Slow Start
 - b) Fast Retransmit
 - c) Fast Recovery
 - d) Tahoe Reset(BL4)(1)
2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:
 - a) 9 MSS
 - b) 16 MSS
 - c) 12 MSS
 - d) 8 MSS(BL4)(1)
3. Which field in the TCP header is used to implement flow control at the receiver side?
 - a) Sequence number
 - b) Acknowledgement number
 - c) Window size
 - d) Urgent pointer(BL4)(1)
4. Nagle's algorithm causes increased latency primarily because it:
 - a) Disables ACK piggybacking
 - b) Buffers small segments until ACK arrives

- c) Reduces CWND on every packet
 - d) Increases SYN backlog
- (BL4)(1)
5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:
- a) MTU size
 - b) CWND growth rate relative to RTT
 - c) DNS resolution time
 - d) UDP competition
- (BL4)(1)
6. UDP is preferred over TCP for DNS queries primarily because:
- a) UDP supports fragmentation
 - b) DNS responses are small and connection overhead is unnecessary
 - c) TCP cannot use port 53
 - d) UDP has better error correction
- (BL4)(1)
7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:
- a) Increase UDP buffer size
 - b) Switch to TCP
 - c) Implement Forward Error Correction (FEC)
 - d) Reduce frame rate to 1 fps
- (BL4)(1)
8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?
- a) 1480 bytes
 - b) 1472 bytes
 - c) 1460 bytes
 - d) 1500 bytes
- (BL4)(1)
9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:
- a) Eliminating IP header overhead
 - b) 0-RTT or 1-RTT connection setup
 - c) Removing encryption
 - d) Native multicast support
- (BL4)(1)
10. HTTP/1.1 persistent connections reduce latency by:
- a) Enabling UDP fallback
 - b) Reusing the same TCP connection for multiple requests
 - c) Compressing HTTP headers
 - d) Sending requests without waiting for DNS
- (BL4)(1)
11. Head-of-line blocking in HTTP/1.1 occurs when:

- a) The DNS server is slow
- b) A response blocks subsequent responses on the same connection
- c) TCP CWND drops to zero
- d) The server rejects pipelined requests

(BL4)(1)

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:

- a) Application layer
- b) DNS layer
- c) TCP layer
- d) IP layer

(BL4)(1)

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?

- a) 301
- b) 302
- c) 304
- d) 307

(BL4)(1)

14. In HTTP caching, the Cache-Control: no-cache directive means:

- a) Never cache this resource
- b) Validate with the server before using a cached copy
- c) Cache for 0 seconds
- d) Disable all caching headers

(BL4)(1)

15. Which DNS record type maps a domain name to an IPv6 address?

- a) A
- b) CNAME
- c) AAAA
- d) MX

(BL4)(1)

16. A DNS resolver switches from UDP to TCP when:

- a) The query type is MX
- b) The response is truncated (TC bit set)
- c) TTL exceeds 3600 s
- d) The authoritative server is unreachable

(BL4)(1)

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?

- a) ARP poisoning
- b) DNS cache poisoning
- c) SYN flooding
- d) BGP hijacking

(BL4)(1)

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:

- a) Cache hit ratio
- b) Recursive resolver query load
- c) DNSSEC signature validity
- d) Zone transfer frequency

(BL4)(1)

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:

- a) HTTP → TCP → DNS
- b) DNS → TCP handshake → HTTP GET
- c) TCP → DNS → HTTP
- d) DNS → HTTP → TCP

(BL4)(1)

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?

- a) TCP for game state, HTTP for assets
- b) UDP with application-layer sequencing for game state, HTTP/3 for assets
- c) HTTP/2 for all traffic
- d) TCP with Nagle enabled for all traffic

(BL4)(1)

Clear-Cut Answers

Q.No.	Correct Answer	Answer
1	c	Fast Recovery
2	b	16 MSS
3	c	Window size
4	b	Buffers small segments until ACK arrives
5	b	CWND growth rate relative to RTT
6	b	DNS responses are small and connection overhead is unnecessary
7	c	Implement Forward Error Correction (FEC)
8	b	1472 bytes
9	b	0-RTT or 1-RTT connection setup
10	b	Reusing the same TCP connection for multiple requests
11	b	A response blocks subsequent responses on the same connection
12	c	TCP layer
13	a	301
14	b	Validate with the server before using a cached copy
15	c	AAAA
16	b	The response is truncated (TC bit set)
17	b	DNS cache poisoning
18	b	Recursive resolver query load
19	b	DNS → TCP handshake → HTTP GET
20	b	UDP with application-layer sequencing for game state, HTTP/3 for assets

Explanations Separately

The following explanations correspond to the 20 questions in the uploaded **CSA07 Computer Networks MCQ Test**.

1. Fast Recovery — Answer: C

Fast Recovery is used by TCP after detecting packet loss through duplicate ACKs. It reduces the congestion window and continues transmission without completely restarting the connection.

2. 9 MSS — Answer: A

The initial CWND is **8 MSS** and SSTHRESH is **16 MSS**. Since CWND is below SSTHRESH, TCP is in Slow Start, so the window increases. For the given MCQ, the expected value is **9 MSS**.

3. Window Size — Answer: C

The **Window Size** field in the TCP header provides receiver-side flow control. It tells the sender how much data the receiver can accept without acknowledgment.

4. Buffers Small Segments — Answer: B

Nagle's algorithm collects small amounts of data and waits for an acknowledgment before sending another small segment. This reduces the number of small packets but can increase latency.

5. CWND Growth Relative to RTT — Answer: B

Satellite links have very large RTTs. Because TCP congestion-window growth depends on acknowledgments, a large RTT makes the window grow more slowly and limits throughput.

6. Small DNS Responses — Answer: B

DNS commonly uses UDP because normal DNS queries and responses are relatively small and do not require establishing a TCP connection first. This reduces communication overhead.

7. Forward Error Correction — Answer: C

With significant packet loss in a streaming application, **FEC** adds redundant information that can help recover lost data without waiting for retransmissions.

8. 1472 Bytes — Answer: B

For an Ethernet MTU of **1500 bytes**:

UDP payload = 1500 – 20 – 8 = 1472 bytes

20 bytes are used by the IPv4 header and 8 bytes by the UDP header.

9. 0-RTT or 1-RTT — Answer: B

QUIC uses UDP and integrates TLS security with the transport protocol. This allows faster connection establishment, including **0-RTT** in suitable cases.

10. Reusing TCP Connection — Answer: B

HTTP/1.1 persistent connections allow multiple HTTP requests and responses to use the same TCP connection. This avoids repeatedly creating new TCP connections.

11. Response Blocking — Answer: B

Head-of-Line blocking occurs when an earlier response is delayed and subsequent responses on the same connection cannot proceed normally until the earlier response is handled.

12. TCP Layer — Answer: C

HTTP/2 solves HOL blocking at the HTTP layer through multiplexing. However, because HTTP/2 normally operates over TCP, packet loss can still cause blocking at the **TCP layer**.

13. HTTP 301 — Answer: A

301 Moved Permanently indicates that a resource has permanently moved to another URL. Browsers and search engines can update their stored location accordingly.

14. Validate Cached Copy — Answer: B

Cache-Control: no-cache does **not** mean that the resource cannot be stored. It means the cached copy should be **validated with the server before being reused**.

15. AAAA — Answer: C

An **AAAA record** maps a domain name to an **IPv6 address**. In comparison, an A record maps a domain name to an IPv4 address.

16. TC Bit Set — Answer: B

If a DNS response received through UDP is too large, the server can set the **Truncated (TC)** bit. The resolver can then retry the query using TCP.

17. DNS Cache Poisoning — Answer: B

DNS cache poisoning involves injecting a forged DNS response into a resolver's cache. A matching transaction ID can help an attacker make a forged response appear legitimate.

18. Recursive Resolver Query Load — Answer: B

A very low TTL means DNS records remain cached for only a short time. They expire more frequently, causing recursive resolvers to make more DNS queries.

19. DNS → TCP → HTTP GET — Answer: B

When visiting a new website, the typical sequence is:

DNS resolution → TCP connection establishment → HTTP GET request

This allows the browser to determine the server address before establishing the connection.

20. UDP + Application Sequencing + HTTP/3 — Answer: B

Real-time multiplayer games require low latency. UDP avoids TCP retransmission delays, while application-layer sequencing can manage game-state packets. HTTP/3 is suitable for transferring web assets efficiently.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions